

Chapter – 2

Savana Online Shopping Database

2.0 Introduction

The entities required for the Savana Online Shopping Database are identified based on the business requirements of the online shopping system. Each entity contains attributes with appropriate primary keys, foreign keys, unique keys, and constraints to maintain data accuracy and consistency. The relationships between entities are identified to represent the interactions within the shopping platform. This analysis serves as the foundation for designing the ER diagram, relational schema, and database implementation while ensuring data integrity and minimizing data redundancy.

2.1 Entity Analysis

Entity: Customer

Attribute	Key
<u>Customer_ID</u>	Primary
First_Name	Not Null
Last_Name	
Email	Unique Key
Phone_Nu	Unique Key
Address	Not Null
Password	Not Null

Entity: Category

Attribute	Key
Category_ID	Primary Key
Category_Na	Unique Key
Description	

Entity: Product

Attribute	Key
Product_ID	Primary Key
Product_Name	Not Null
Category_ID	Foreign Key
Price	Not Null
Stock	Check
Brand	
Color	
Size	

Entity: Cart

Attribute	Key
Cart_ID	Primary Key

Customer_ID	Foreign Key
Total_Items	
Grand_Total	

Entity: Order

Attribute	Key
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	
Order_Status	
Total_Amount	

Entity: Order_Item

Attribute	Key
OrderItem_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	
Unit_Price	

Entity: Payment

Attribute	Key
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	
Payment_Status	
Payment_Date	
Amount	

Entity: Shipment

Attribute	Key
Shipment_ID	Primary Key
Order_ID	Foreign Key
Delivery_Address	
Shipment_Date	
Delivery_Date	
Shipment_Status	

Entity: Review

Attribute	Key
Review_ID	Primary Key
Customer_ID	Foreign Key
Product_ID	Foreign Key
Rating	Check
Review_Text	
Review_Date	

2.2 Entity Relationships

Entity 1	Relationship	Entity 2	Cardinality	Participation
Customer	Owns	Cart	One-to-One	Partial / Total
Customer	Places	Order	One-to-Many	Partial / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Category	Contains	Product	One-to-Many	Total / Total
Cart	Contains	Product	Many-to-Many	Partial / Partial
Order	Includes	Order_Item	One-to-Many	Total / Total
Product	Appears In	Order_Item	One-to-Many	Partial / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Shipment	One-to-One	Total / Total
Product	Receives	Review	One-to-Many	Partial / Total